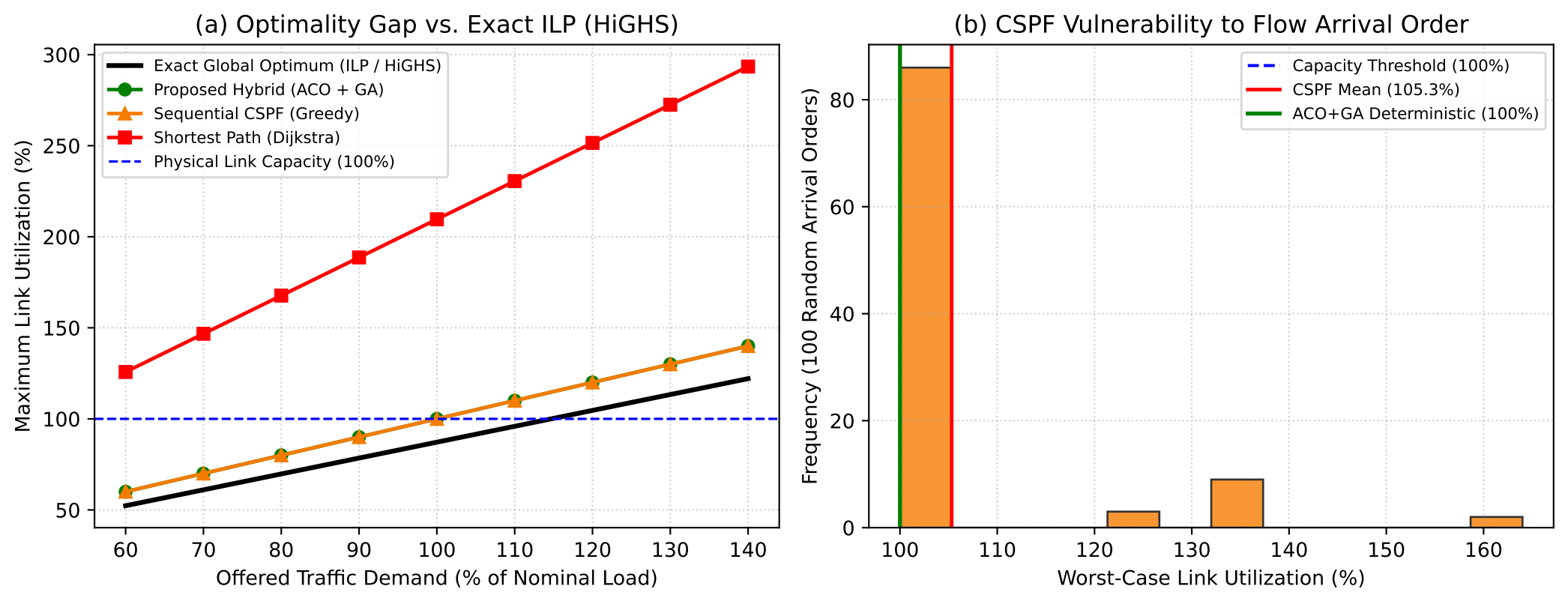
(a) Optimality Gap vs. Exact ILP (HiGHS)
Exact Global Optimum (ILP / HiGHS)
Proposed Hybrid (ACO + GA)
Sequential CSPF (Greedy)
Shortest Path (Dijkstra)
Physical Link Capacity (100%)
Maximum Link Utilization (%)
Offered Traffic Demand (% of Nominal Load)
300
250
200
150
100
50
60
70
80
90
100
110
120
130
140
(b) CSPF Vulnerability to Flow Arrival Order
Capacity Threshold (100%)
CSPF Mean (105.3%)
ACO+GA Deterministic (100%)
Frequency (100 Random Arrival Orders)
Worst-Case Link Utilization (%)
80
60
40
20
0
100
110
120
130
140
150
160